Table 1: Phase 1 matched-compute contrasts under Protocol A at 16K context. Differences are MV minus the reference, in points, so positive values favour MV. Each item is scored as its mean over the three decoding seeds, and the 95% intervals come from the paired item-level bootstrap of Appendix **??**, with items resampled within each benchmark. This table uses 20,000 replicates rather than 2,000, so that its q-values are not limited by the smallest p-value the bootstrap can give. q is the Benjamini–Hochberg adjusted q-value, computed within each column for the per-task rows, over the four suite-average rows, and over the ten reweighted rows. The reweighted rows recompute the suite average after SFT with the four domains of Table **??** weighted equally, or with one domain left out, as Table **??** does for the front-loading study. n is the number of scored items: HumanEval counts its 164 Python problems and IFEval the 769 instructions of its 500 prompts. A dash marks a benchmark on which both models score zero on every item, which is not tested.

	MV − Comma0.1			MV − Nemotron-CC-HQ		
Task	Δ pp	95% CI	q	Δ pp	95% CI	q
Per task, after Tulu3 SFT						
AIME24 (n=30)	+3.3	[+0.0, +10.0]	0.793	+3.3	[+0.0, +10.0]	0.792
AIME25 (n=30)	+3.3	[+0.0, +10.0]	0.793	+3.3	[+0.0, +10.0]	0.792
AMC23 (n=40)	+12.5	[+2.5, +22.5]	0.017	+15.0	[+5.0, +27.5]	0.003
MATH500 (n=500)	+32.5	[+28.3, +36.7]	< 0.001	+32.1	[+28.0, +36.1]	< 0.001
GSM8K (n=1319)	+44.7	[+41.9, +47.5]	< 0.001	+43.0	[+40.1, +45.9]	< 0.001
HumanEval (n=164)	+37.6	[+29.9, +45.3]	< 0.001	+48.8	[+41.1, +56.7]	< 0.001
LiveCodeBench (n=369)	+9.5	[+6.6, +12.6]	< 0.001	+9.7	[+6.8, +12.7]	< 0.001
MBPP (n=500)	+15.2	[+11.0, +19.5]	< 0.001	+34.5	[+30.2, +38.9]	< 0.001
GPQA-Diamond (n=198)	+0.2	[−7.9, +8.2]	0.971	+1.2	[−7.2, +9.6]	0.792
JEEBench (n=515)	+0.3	[−1.3, +2.0]	0.793	+3.9	[+2.4, +5.6]	< 0.001
IFEval (n=769)	+4.7	[+0.7, +8.6]	0.030	+9.0	[+5.3, +12.7]	< 0.001
Eleven-task suite average						
Tulu3 SFT, 16K	+14.9	[+13.2, +16.7]	< 0.001	+18.5	[+16.8, +20.4]	< 0.001
SFT→DPO, 16K	+15.0	[+13.3, +16.7]	< 0.001	+18.6	[+16.8, +20.4]	< 0.001
Suite average reweighted, after Tulu3 SFT						
Equal-domain average	+11.2	[+9.5, +13.0]	< 0.001	+15.5	[+13.7, +17.3]	< 0.001
Drop math domain	+11.2	[+9.1, +13.4]	< 0.001	+17.9	[+15.7, +20.0]	< 0.001
Drop code domain	+12.7	[+10.6, +15.0]	< 0.001	+13.9	[+11.8, +16.2]	< 0.001
Drop science domain	+18.2	[+16.3, +20.2]	< 0.001	+22.1	[+20.2, +24.2]	< 0.001
Drop instruction domain	+15.9	[+14.1, +17.9]	< 0.001	+19.5	[+17.6, +21.5]	< 0.001

Table 2: Statistics for the 100B mixture-composition ablation of Section **??**, under Protocol A like Table **??**. Differences are the ablation minus the full 100B mixture (MV-100), in points, so negative values mean the ablation is worse. The method is that of Table 1: each item is scored as its mean over the three decoding seeds, and the 95% intervals come from the paired item-level bootstrap of Appendix **??**, with items resampled within each benchmark and 20,000 replicates. q is the Benjamini–Hochberg adjusted q-value, computed within each column for the per-task rows and over the twelve suite-average rows. n is the number of scored items: HumanEval counts its 164 Python problems and IFEval the 769 instructions of its 500 prompts. A dash marks a benchmark on which both models score zero on every item, which is not tested.

	MV-noweb − MV-100			MV-noinstruct − MV-100		
Setting or task	Δ pp	95% CI	q	Δ pp	95% CI	q
Eleven-task suite average						
Base, 4K	+10.0	[+8.6, +11.4]	< 0.001	−3.7	[−4.7, −2.7]	< 0.001
Tulu3 SFT, 4K	+1.2	[+0.0, +2.5]	0.062	−9.0	[−10.3, −7.8]	< 0.001
SFT→DPO, 4K	+1.5	[+0.1, +3.0]	0.048	−9.1	[−10.5, −7.7]	< 0.001
Base, 16K	+0.1	[−1.4, +1.6]	0.933	−12.9	[−14.3, −11.5]	< 0.001
Tulu3 SFT, 16K	+2.5	[+1.2, +3.9]	< 0.001	−7.4	[−8.6, −6.3]	< 0.001
SFT→DPO, 16K	+2.7	[+1.2, +4.2]	< 0.001	−7.7	[−8.8, −6.5]	< 0.001
Per task, after Tulu3 SFT at 16K						
AIME24 (n=30)	+3.3	[+0.0, +10.0]	0.916	—	—	—
AIME25 (n=30)	—	—	—	—	—	—
AMC23 (n=40)	+7.5	[+0.0, +17.5]	0.210	—	—	—
MATH500 (n=500)	+17.5	[+13.8, +21.4]	< 0.001	−3.5	[−5.5, −1.5]	< 0.001
GSM8K (n=1319)	−3.2	[−6.0, −0.3]	0.155	−32.1	[−34.7, −29.4]	< 0.001
HumanEval (n=164)	+2.8	[−3.9, +9.6]	0.838	−23.4	[−30.5, −16.5]	< 0.001
LiveCodeBench (n=369)	+1.4	[+0.0, +3.0]	0.170	−0.3	[−1.1, +0.4]	0.677
MBPP (n=500)	−0.1	[−3.4, +3.1]	0.942	−18.9	[−23.1, −14.9]	< 0.001
GPQA-Diamond (n=198)	−1.2	[−7.1, +4.7]	0.916	+1.7	[−6.7, +9.9]	0.713
JEEBench (n=515)	−0.5	[−2.5, +1.4]	0.916	−2.1	[−3.9, −0.3]	0.036
IFEval (n=769)	−0.3	[−3.9, +3.4]	0.942	−3.4	[−7.0, +0.3]	0.102